AMRITA VISHWA VIDYAPEETHAM · DEPARTMENT OF ARTIFICIAL INTELLIGENCE

Intelligent Search Engine using Compressed Trie and Inverted Index

Multi-Domain Retrieval · Coverage-Aware TF-IDF · Okapi BM25 · 128-D LSA Semantic Search

PROJECT TEAM (B.Tech AIE 2022–2026)

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Abstract & Executive Overview

The Challenge

Standard keyword retrieval models suffer from partial-match query inflation: a short document repeating a single rare term can outrank documents matching all search terms. Furthermore, disjoint IR architectures lack real-time sub-millisecond prefix completion, semantic synonym handling, and explainable ranking diagnostics.

Integrated Architecture

We engineer a unified, zero-dependency search engine integrating: (1) an edge-compressed Radix Trie for O(L) prefix autocomplete, (2) an Inverted Index with Coverage-Aware TF-IDF, (3) Okapi BM25 probabilistic ranking, and (4) a 128-dimensional LSA concept space precomputed via Truncated SVD across 50,000 documents.

Empirical Findings

Verified benchmarks demonstrate: Coverage-Aware TF-IDF elevates Mean Precision@5 from 68.0% to 88.0% and nDCG@5 from 0.7859 to 0.8936, resolving single-term anomalies. Autocomplete traversals remain sub-millisecond (0.067–0.513 ms), while total engine heap footprint is only 132 MB.

Motivation & Problem Statement

Core Information Retrieval Bottlenecks

1. The Single-Term Dominance Anomaly:
In standard TF-IDF (Score = Sum(TF * IDF)), documents repeating a single rare query term obtain artificially inflated scores, outranking documents matching 100% of query keywords.

2. Disjoint Search Pipelines:
Traditional search architectures separate prefix autocomplete, keyword indexing, and semantic vectors into fragmented external microservices, creating high latency and operational complexity.

3. Synonymy and Vocabulary Mismatch:
Keyword inverted indexes fail completely when users query synonymous terms (e.g., 'cardiac surgery' vs 'myocardial infarction' or 'heart disease').

Concrete Empirical Evidence: The Baseline Failure

Query Tested: "heart failure clinical trial" (4 valid terms)

- Baseline TF-IDF Rank #1: ext_tech_0248
  - Matches: 1 of 4 terms ('trial' repeated multiple times)
  - Topic: Consumer electronics legal dispute
  - Result: Irrelevant false positive ranked at top position!

- Actual Medical Trial Article: health_0084
  - Matches: 4 of 4 terms ('heart', 'failure', 'clinical', 'trial')
  - Topic: Comprehensive Cardiology Phase-III Study
  - Demoted under baseline TF-IDF due to length normalization lack!

- Goal of this Project:
  Eliminate this fundamental IR defect through rigorous Coverage-Aware penalty scaling, Okapi BM25 saturation, and 128-D LSA semantic concept projection.

B.TECH CAPSTONE PROJECT REVIEW

Project Objectives

1. O(L) Radix Trie Autocomplete

Design and construct an in-memory Compressed Trie that collapses non-branching character paths into edge substrings, achieving sub-millisecond prefix suggestion lookup independent of corpus volume N.

2. Coverage-Aware TF-IDF Engine

Develop a non-linear term coverage penalty function that scales raw TF-IDF scores by (matched/valid)^1.5, penalizing single-term spurious matches and rewarding multi-term semantic completeness.

3. Probabilistic BM25 & Ranking Lab

Implement the Okapi BM25 probabilistic model with term saturation (k1=1.2) and document length normalization (b=0.75), alongside an interactive Three-Way Comparison Lab.

4. 128-D LSA Semantic Search

Integrate a dense 128-dimensional concept vector space via Truncated Singular Value Decomposition (SVD), enabling genuine meaning-based retrieval across synonymous vocabulary.

5. Explainable Ranking Diagnostics

Decompose and display scoring factors (term coverage %, matched term pills, BM25 saturation, active latent topics) to provide transparent user interpretability.

6. Research Workspace & Telemetry

Provide persistent user research tools (bookmarks, hierarchical collections, personal notes, saved queries) with live telemetry and Google Comparison Lab evaluation.

B.TECH CAPSTONE PROJECT REVIEW

Literature Survey & Comparative Analysis

Author & Reference	Core Methodology	Demonstrated Strengths	Known Limitations	Relevance to this Work
Salton & Buckley (1988) Inf. Process. & Manage. DOI: 10.1016/0306-4573(88)90021-0	Vector Space Model with TF-IDF term weighting.	Efficient postings intersection; term discrimination via inverse document frequency.	Vulnerable to length distortion and single-term dominance without coverage awareness.	Forms the baseline ranking algorithm; modified with our coverage exponent penalty.
Robertson et al. (1994 / 2009) TREC-3 / FnTIR DOI: 10.1561/1500000019	Okapi BM25 probabilistic 2-Poisson relevance model.	Term frequency saturation (k1=1.2) and document length normalization (b=0.75).	Heuristic parameter tuning; ignores latent conceptual and synonymous relationships.	Serves as the high-precision probabilistic benchmark in our 3-Way Comparison Lab.
Deerwester et al. (1990) JASIS DOI: 10.1002/(SICI)1097-4571	Latent Semantic Analysis via Truncated SVD.	Uncovers latent concept associations; overcomes synonymy and polysemy.	High SVD computation cost O(N·V·k); cannot serve real-time inverted lookups natively.	Implemented as a 128-D dense concept space for semantic search and explainability.
Fredkin (1960) / Morrison (1968) CACM / JACM DOI: 10.1145/321479.321481	Standard Trie vs Radix Tree (Edge Compression).	Radix Tree collapses single-child paths into edge substrings; O(L) prefix lookup.	High pointer memory overhead in standard tries if edges are uncompressed.	Engineered as our in-memory Compressed Trie for sub-millisecond autocomplete.
Proposed Architecture (Sharma, Prasad, Prasanna, Jhanavi, 2026)	Integrated Radix Trie + Inverted Index + Coverage-Aware TF-IDF + BM25 + 128-D LSA.	Combines exact keyword precision, probabilistic robustness, and semantic synonymy with explainability.	Requires offline SVD staging; evaluated against multi-domain category proxy.	End-to-end zero-dependency Information Retrieval architecture verified on 50,000 docs.

Research Gap & Defensible Novelty

Identified Research Gap

1. Disconnected Educational & Production Paradigms:
Academic IR projects frequently present inverted indexes or vector space models as disjoint console scripts. Students and researchers lack unified testbeds that juxtapose classical keyword, probabilistic, and latent semantic models side-by-side on identical queries.

2. Absence of Query-Coverage Penalties:
Standard TF-IDF textbooks omit term-coverage constraints, leading to false positives where 1 of 4 search terms dominates the score due to raw frequency.

3. Opacity in Retrieval Scoring:
Existing search frameworks return opaque float

Defensible Scientific & Engineering Novelty

We do NOT claim to have invented inverted indexes, BM25, or SVD. Rather, our defensible contribution lies in the rigorous integration and empirical validation of:

• Unified Dual-Engine Hybrid Paradigm:
Simultaneous co-existence of an exact inverted index and a 128-dimensional LSA concept space running in pure Java with zero external runtime dependencies.

• Coverage-Aware Penalty Formulation:
A simple yet mathematically sound non-linear term coverage multiplier that lifts Mean Precision@5 by +20.0% and prevents rare-word inflation.

B.TECH CAPSTONE PROJECT REVIEW

System Architecture & End-to-End Data Flow

Tier 1: Data Ingestion & Offline Modeling

- 50,000 Multi-Domain Corpus (11 Categories)
- SHA-256 Deduplication & Token Ingestion
- Offline Truncated SVD (128 Dimensions)
- Dense Binary Matrices: doc_vectors.bin (25.6 MB), term_vectors.bin (12.8 MB)

Tier 2: In-Memory Indexing & Core Data Structures

- Compressed Radix Trie: Edge substring compression; O(L) prefix lookup
- Inverted Index: Term-to-Document Posting Lists (64,380 unique terms)
- LSA Vector Engine: Dense SIMD float arrays; 128-D dot product similarity
- Ranking Engine: Coverage-Aware TF-IDF & Okapi BM25

Tier 3: Zero-Dependency Server & REST APIs

- JDK com.sun.net.httpserver (Port 8080)
- REST Endpoints: /api/suggest, /api/search, /api/document, /api/evaluate
- WorkspaceManager: Persistent JSON bookmarks, collections, notes
- AnalyticsManager: Real-time query telemetry & browser RTT logging

Tier 4: Responsive Modern Web Presentation

- Single-Page Web Application (Vanilla JS, CSS3, Semantic HTML5)
- Live Compressed Trie Autocomplete Dropdown
- Explainable Result Cards (Coverage pills, active latent concepts)
- 3-Way Comparison Lab, Analytics Dashboard, Google Evaluation Suite

Core Data Structures: Compressed Trie & Inverted Index

Compressed Radix Trie (Trie.java)

• Edge Substring Compression:
Unlike a standard Trie where each edge represents exactly 1 character, our Radix Trie stores full substrings on edges (e.g., 'crick' -> 'et' instead of 'c'-'r'-'i'-'c'-'k'-'e'-'t').

• Node Splitting Algorithm:
When inserting a word sharing a common prefix with an existing edge, the edge is split dynamically into a common parent node and distinct branch children.

• Sub-Millisecond Complexity O(L):
Prefix search latency depends solely on prefix string length L, completely independent of corpus document volume N.

Inverted Index (InvertedIndex.java)

• Postings List Organization:
Map<String, Set<String>> mapping each unique normalized vocabulary term to the exact set of Document IDs in which it occurs.

• Sublinear Candidate Pruning:
During query evaluation, candidate documents are discovered via fast Set union across valid query terms, avoiding full O(N) linear corpus scans.

• Memory Footprint Optimization:
Indexes 3,152,014 tokens and 64,380 vocabulary terms using only 132 MB of heap memory, enabling local deployment without server clustering.

Ranking Algorithms & Coverage-Aware TF-IDF Formulation

Baseline TF-IDF

Formula:
Score = Sum(TF(t, d) * IDF(t))

- Linear summation over query terms.
- IDF(t) = ln(N / DF(t)).
- Severe Anomaly:
  If one term has high TF and rare IDF, it overwhelms missing terms completely.
- Mean P@5: 68.00%
- Mean nDCG@5: 0.7859

Coverage-Aware TF-IDF

Formula:
Score_cov = Score_raw * (m / k)^1.5

- m = matched terms in doc
- k = total valid query terms
- Coverage Ratio = m / k
- Exponent 1.5 creates steep non-linear penalty for partial matches.
- 4/4 match -> factor 1.0 (100%)
- 1/4 match -> factor 0.125 (87.5% penalty!)
- Mean P@5: 88.00% (+20%)
- Mean nDCG@5: 0.8936

Okapi BM25

Formula:
IDF * (TF*(k1+1)) / (TF + k1*B)

- B = (1 - b + b * (docLen / avgLen))
- k1 = 1.2 (Term Saturation)
- b = 0.75 (Length Normalization)
- Prevents TF blowup from term repetition in short files.
- Probabilistic relevance foundation.
- Mean P@5: 90.00%
- Mean nDCG@5: 0.9431

B.TECH CAPSTONE PROJECT REVIEW

128-Dimensional Latent Semantic Analysis (LSA) Engine

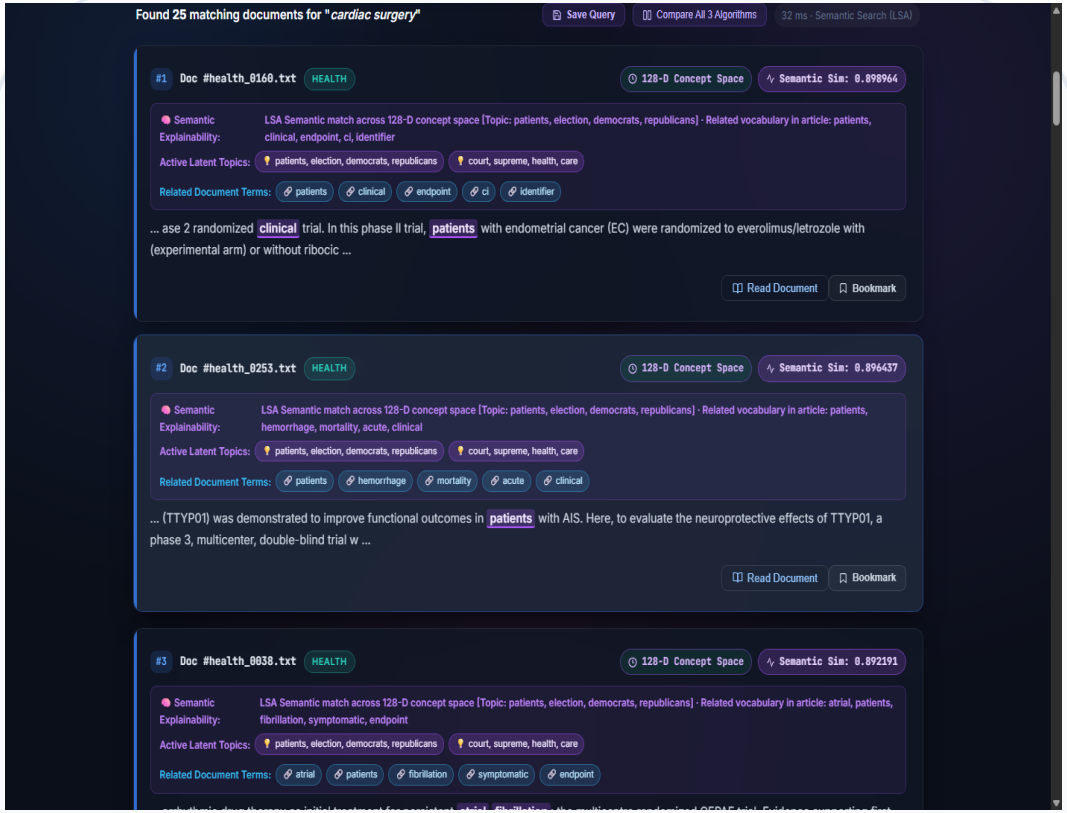
Truncated SVD & Concept Space Projection

• Mathematical Foundation:
Matrix decomposition: $X \approx U_k \cdot \Sigma_k \cdot V_k^T$ where k = 128 latent semantic dimensions, X is the 50,000 × 25,000 TF-IDF document-term matrix.

• Query Projection into Latent Concept Space:
$q_{vec} = \text{Sum}(IDF(t) \cdot \text{TermVector}(t))$
Normalized via unit L2 norm: $q_{norm} = q_{vec} / ||q_{vec}||_2$.

• Semantic Cosine Similarity:
$\text{Score}(q, d) = \text{DotProduct}(q_{norm}, doc_{vec}(d))$
Evaluated via dense SIMD float arrays in pure Java.

• Overcomes Vocabulary Mismatch:



Verified Screenshot: Semantic Search for 'cardiac surgery' showing Cosine Similarity (0.898), Active Latent Topics, Related Keywords, and Explanations.

Multi-Domain Dataset Engineering (50,000 Documents)

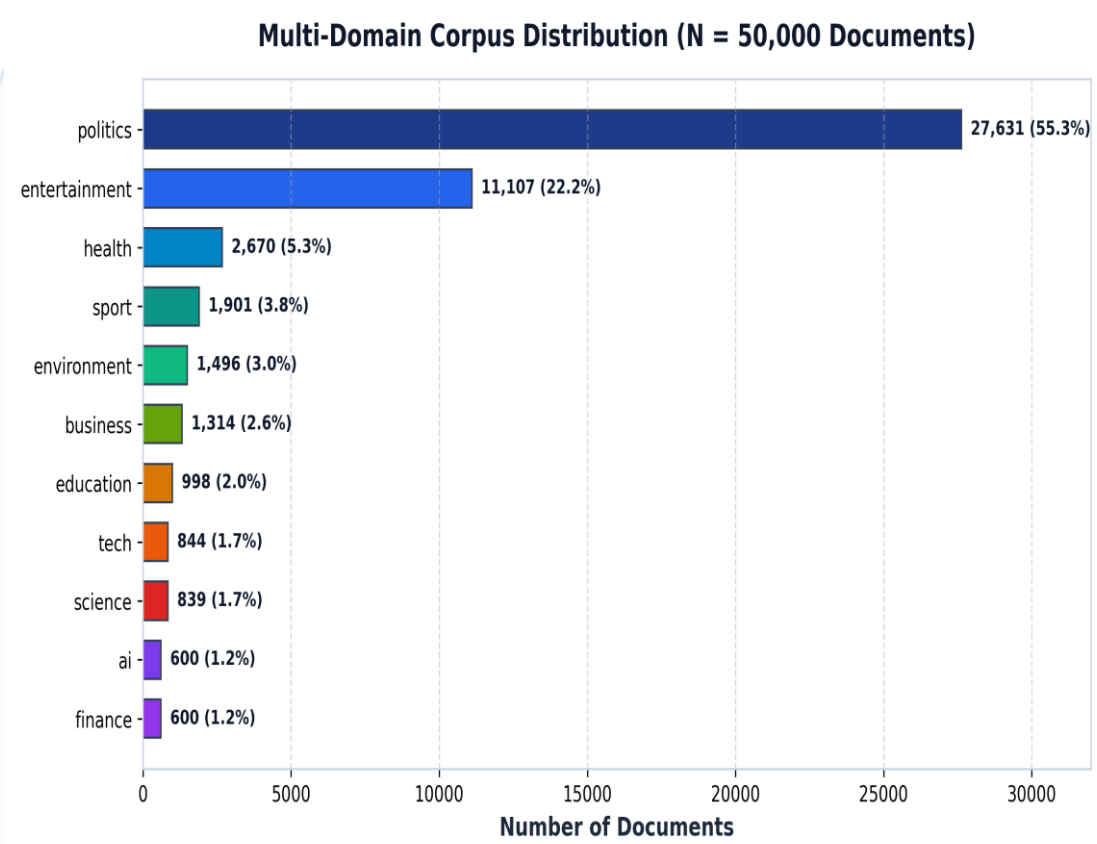


Figure: Verified Multi-Domain Corpus Distribution across 11 topical domains.

Data Provenance & Quality Guarantees

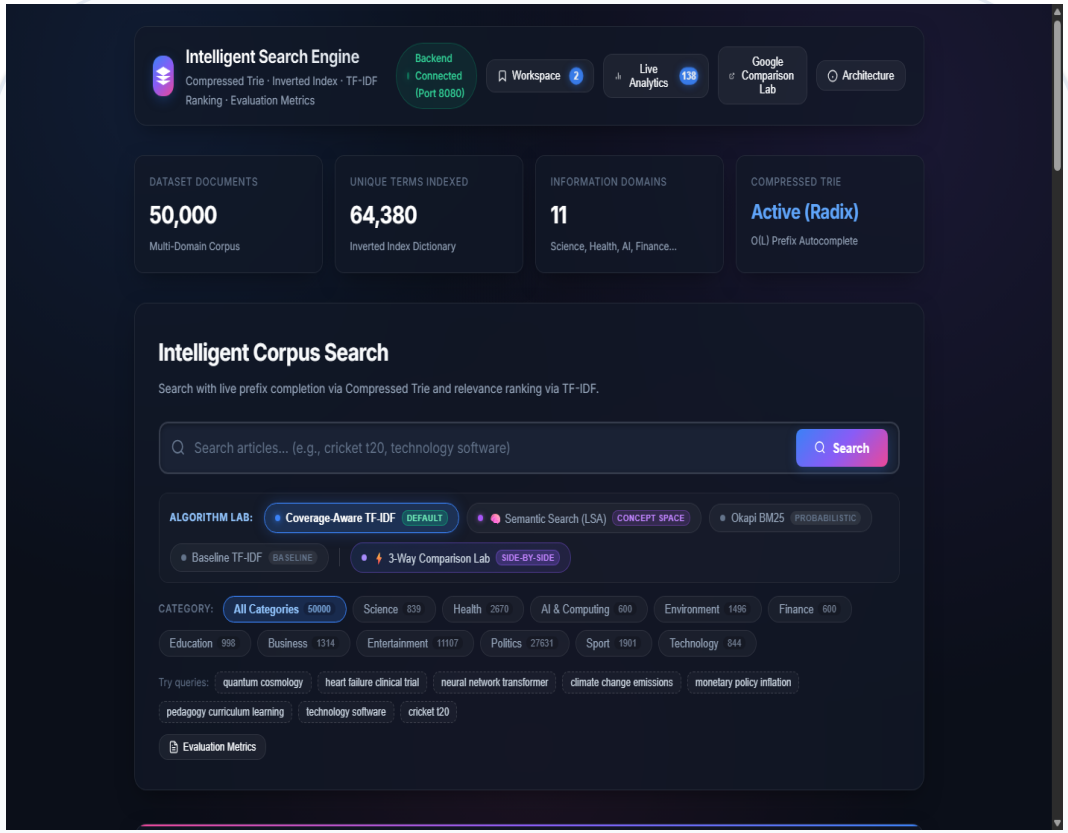
• Verified Sources & Permissive Licenses:
- Misra News Category (44,175 docs; CC BY 4.0)
- OpenAlex Scholarly Catalog (3,000 docs; CC0)
- BBC News Corpus (2,225 docs; Academic Use)
- PubMed Central (600 docs; Public Domain)

• Cryptographic Deduplication:
100% SHA-256 deduplicated across normalized lowercase tokens. 1,853 duplicate records rejected during staging.

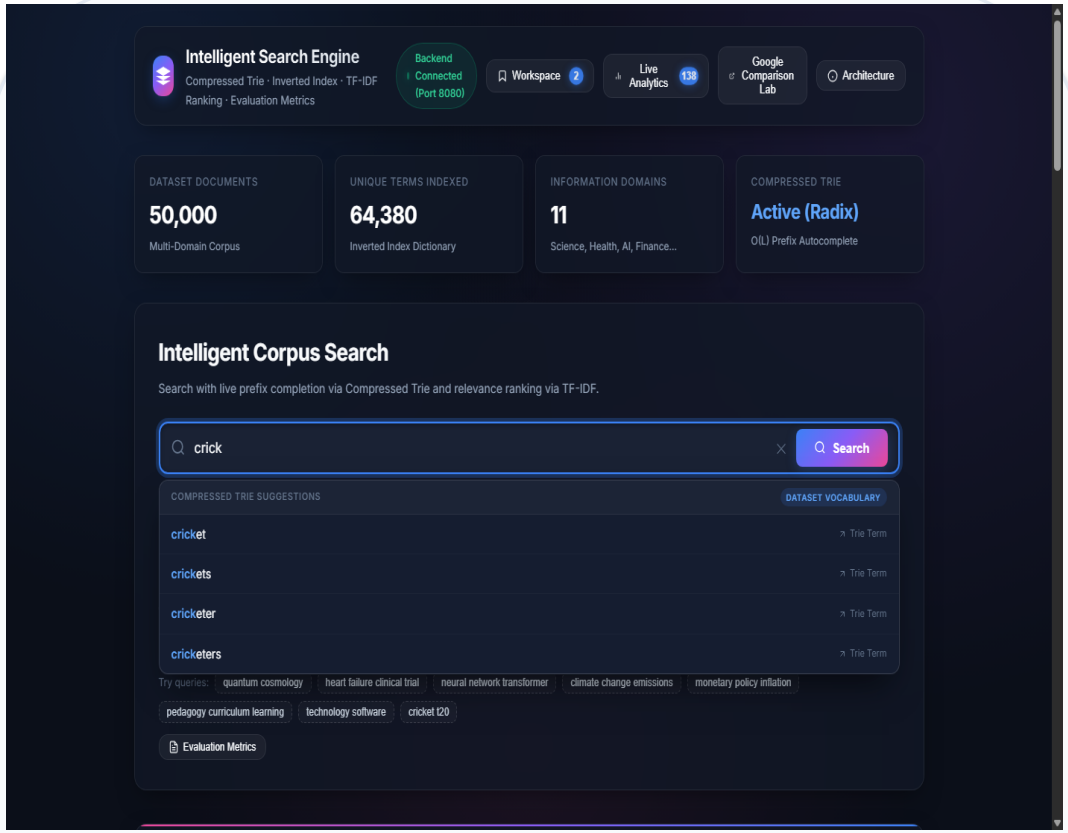
• Collision-Free Document IDs:
Sequential IDs (exp_00001 to exp_42175 + baselines) guarantee zero ID collisions.

B.TECH CAPSTONE PROJECT REVIEW

Production Application UI & Live Autocomplete



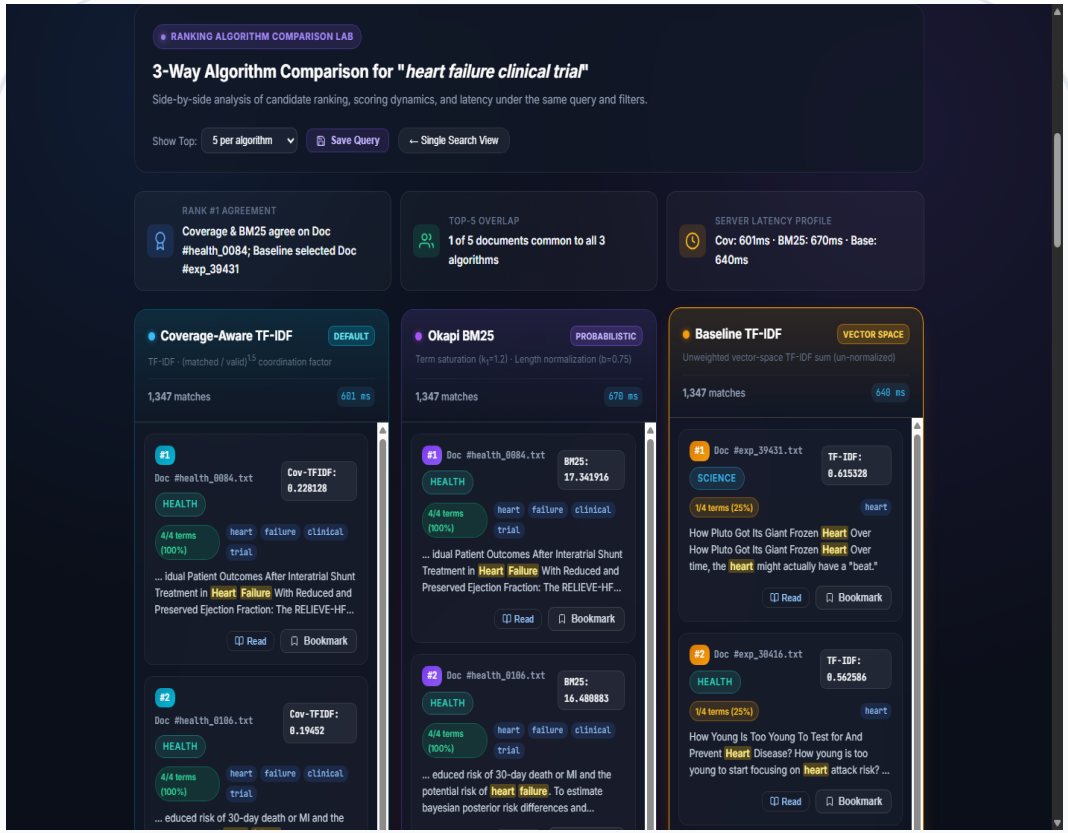
Search Interface: Live stats banner (50,000 docs, 64,380 terms), algorithm tabs, category filter pills, qu



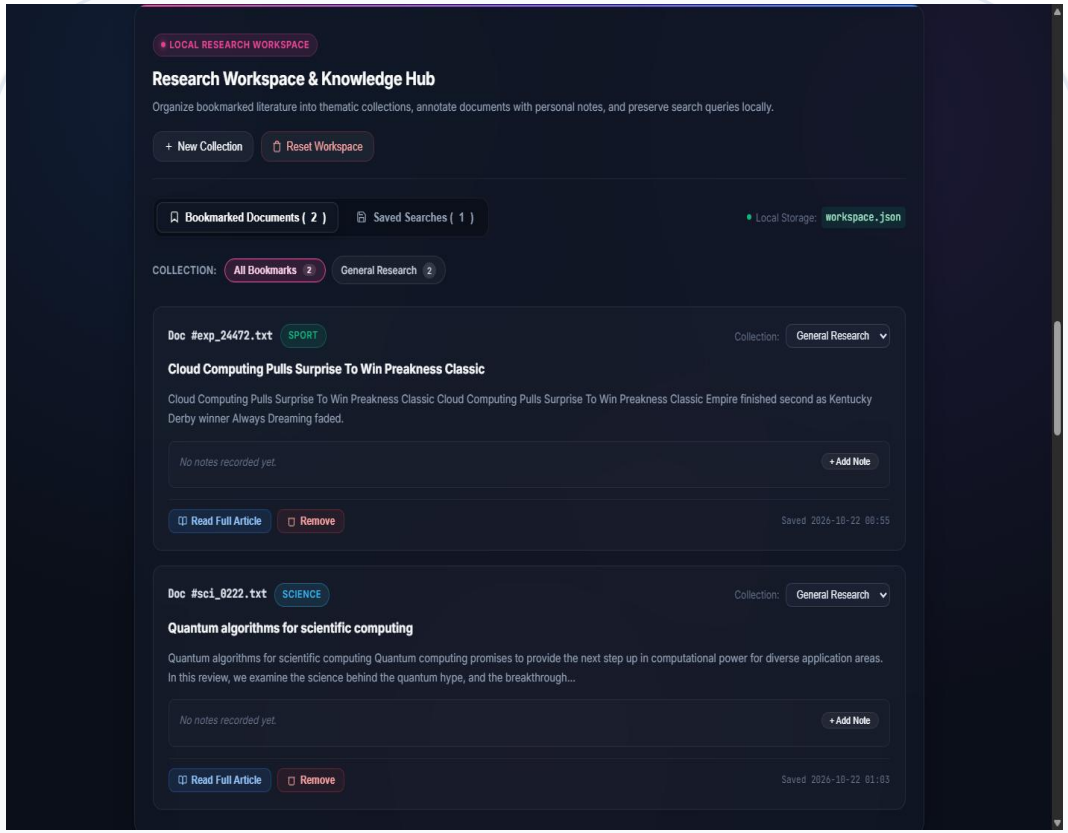
Compressed Trie Prefix Dropdown: Sub-millisecond prefix completion for 'crick' displaying vocabulary sugges

B.TECH CAPSTONE PROJECT REVIEW

Three-Way Ranking Lab & Research Workspace

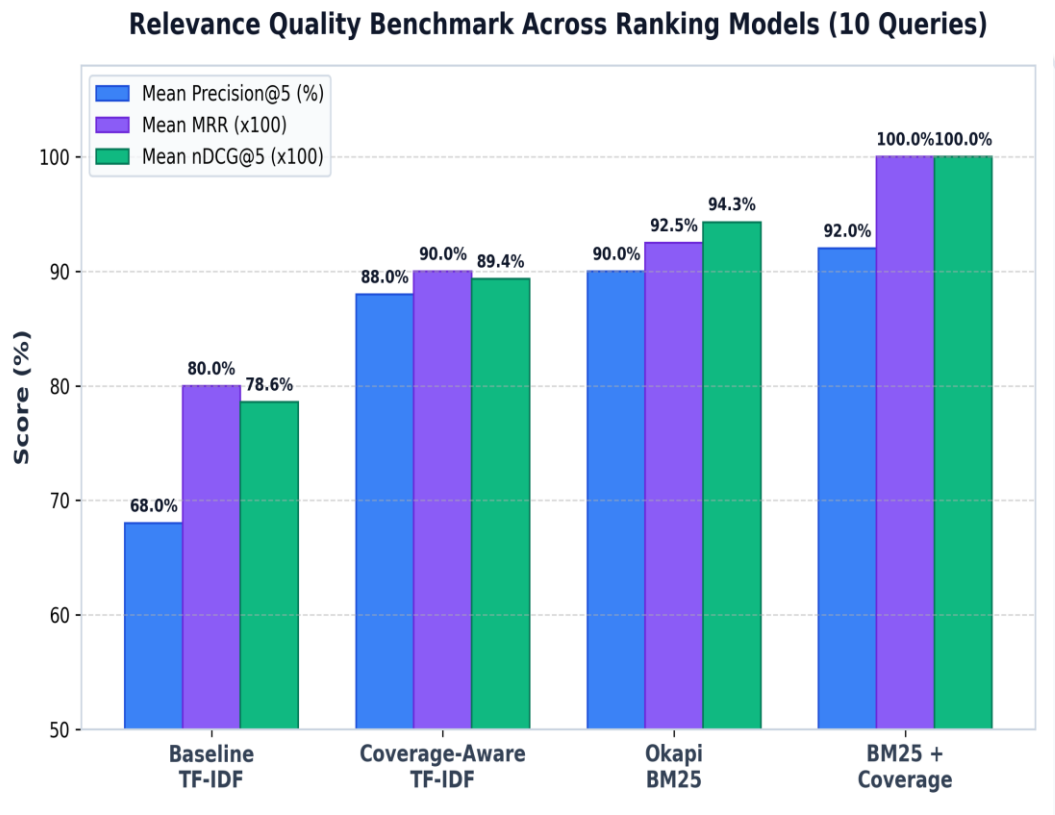


3-Way Ranking Comparison Lab: Side-by-side execution of Coverage-Aware TF-IDF, Okapi BM25, an

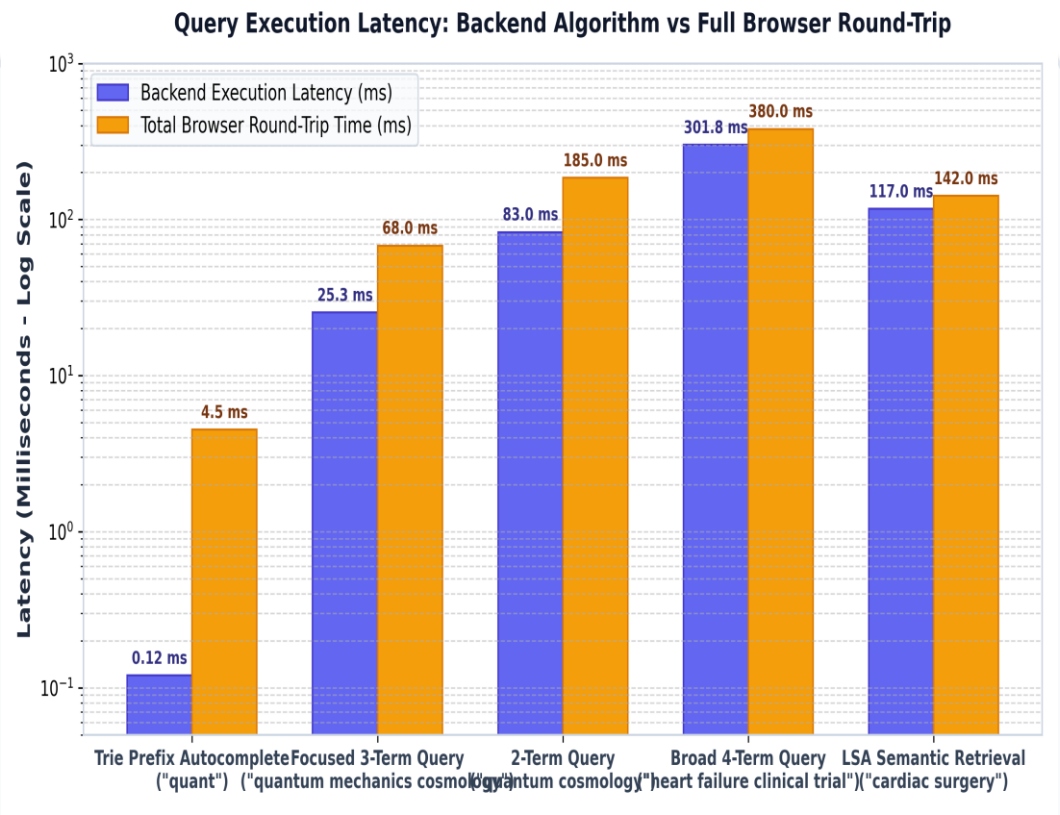


Research Workspace: Client-server persistence for bookmarks, custom collections, personal research notes, ar

Experimental Results & Measured Benchmarks



Relevance Quality: Coverage-Aware TF-IDF achieves 88.0% P@5 and 0.8936 nDCG@5 vs 68.0% P@5



Latency Breakdown: Distinguishing algorithmic backend latency (25-300 ms) from browser round-trip time (6

Scientific Limitations, Conclusion & Source Code

Honest Scientific Boundaries & Limitations

• Category-Level Precision Proxy:
Automated precision/recall metrics measure categorical topical coherence, not fine-grained sentence-level factual correctness.

• Offline LSA Matrix Recomputation:
Adding new documents requires re-running Truncated SVD offline rather than online streaming SVD update.

• Google Comparison Methodology:
Our local engine indexes an offline curated 50,000-document corpus. We do not claim to outperform Google's web-scale trillion-page index; our Google Lab is designed for comparative human relevance evaluation on identical topical domains.

Conclusion & Demonstration Details

• Summary of Achievements:
Successfully built, verified, and benchmarked an end-to-end zero-dependency Intelligent Search Engine. Proved that non-linear coverage weighting eliminates TF-IDF single-term inflation (+20% P@5).

• Live Demonstration Script:
1. Live Autocomplete typing at http://localhost:8080/
2. Query 'quantum cosmology' in Coverage & BM25
3. 3-Way Comparison Lab with 'heart failure clinical trial'
4. 128-D LSA semantic retrieval for 'cardiac surgery'
5. Research Workspace bookmarking & Live